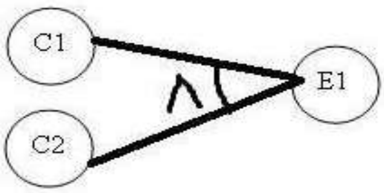
Cause and Effect Graph Technique

Cause and effect graph is a dynamic test case writing technique where causes are the input conditions and effects are the results of those input conditions. Cause-Effect Graphing is a technique which starts with set of requirements and determines the minimum possible test cases for maximum test coverage which reduces test execution time and ultimately cost.

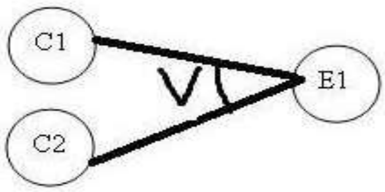
The goal is to reduce the total number of test cases still achieving the desired application quality by covering the necessary test cases for maximum coverage. But at the same time obviously there are some downsides of using this test case writing technique. It takes time to model all your requirements into this cause-effect graph before writing test cases.

The Cause-Effect graph technique restates the requirements specification in terms of logical relationship between the input and output conditions. Since it is logical, it is obvious to use Boolean operators like AND, OR and NOT. Notation are shown belo

AND – For effect E1 to be true, both the causes C1 and C2 should be true

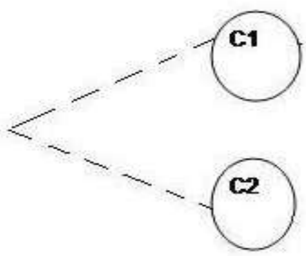


OR – For effect E1 to be true, either of causes C1 OR C2 should be true



NOT – For Effect E1 to be True, Cause C1 should be false

MUTUALLY EXCLUSIVE – When only one of the causes will hold true.



<u>Now let’s try to implement this technique with some example.</u>

1. Draw a cause and effect graph based on a requirement/situation
2. Cause and Effect graph is given, draw a decision table based on it to draw the test case.

Let’s see both of them one by one.

Let’s draw a cause and effect graph based on a situation

***Situation*:**

The “Print message” is software that read two characters and, depending of their values, messages must be printed.

- The first character must be an “A” or a “B”.
- The second character must be a digit.
- If the first character is an “A” or “B” and the second character is a digit, the file must be updated.
- If the first character is incorrect (not an “A” or “B”), the message X must be printed.
- If the second character is incorrect (not a digit), the message Y must be printed.

Solution:

The causes for this situation are:

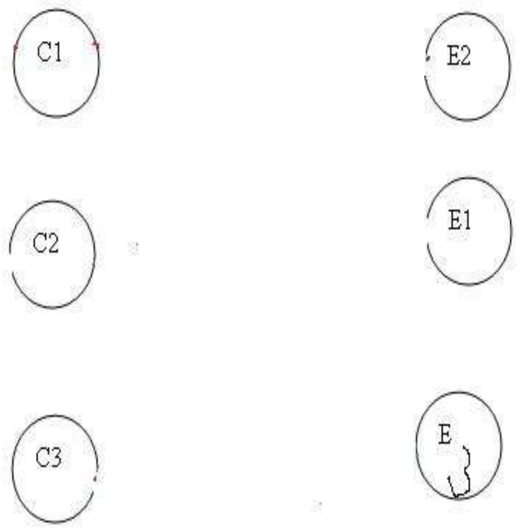
C1 – First character is A
C2 – First character is B
C3 – Second character is a digit

The effects (results) for this situation are

E1 – Update the file
E2 – Print message “X”
E3 – Print message “Y”

LET'S START!!

First draw the causes and effects as shown below:



Key – Always go from effect to cause (left to right). That means, to get effect "E" ,what causes should be true.

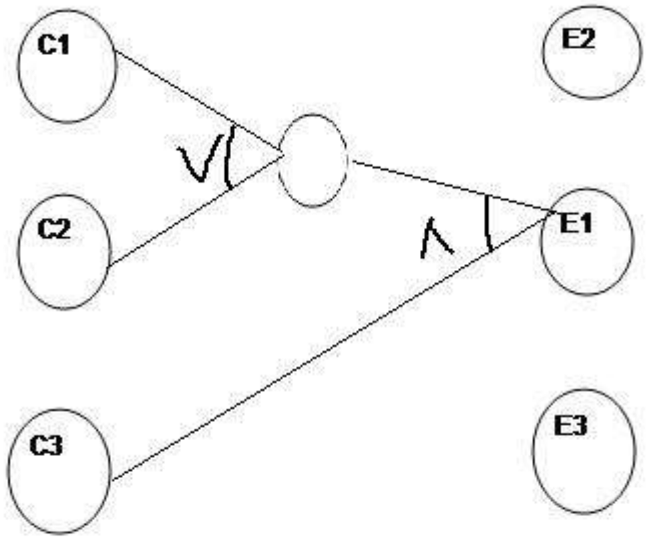
In this example, let's start with Effect E1.

Effect E1 is to update the file. The file is updated when
– First character is "A" and second character is a digit
– First character is "B" and second character is a digit
– First character can either be "A" or "B" and cannot be both.

Now let's put these 3 points in symbolic form:

For E1 to be true – following are the causes:
– C1 and C3 should be true
– C2 and C3 should be true
– C1 and C2 cannot be true together. This means C1 and C2 are mutually exclusive.

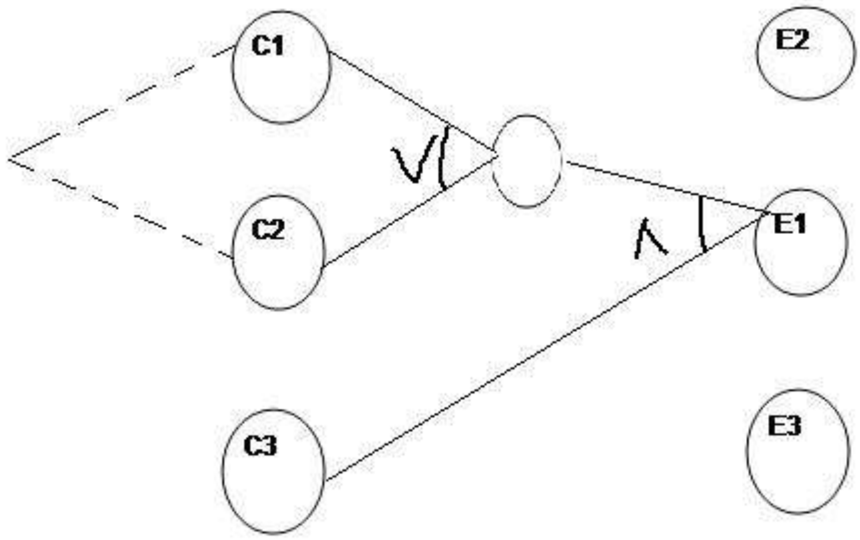
Now let's draw this:



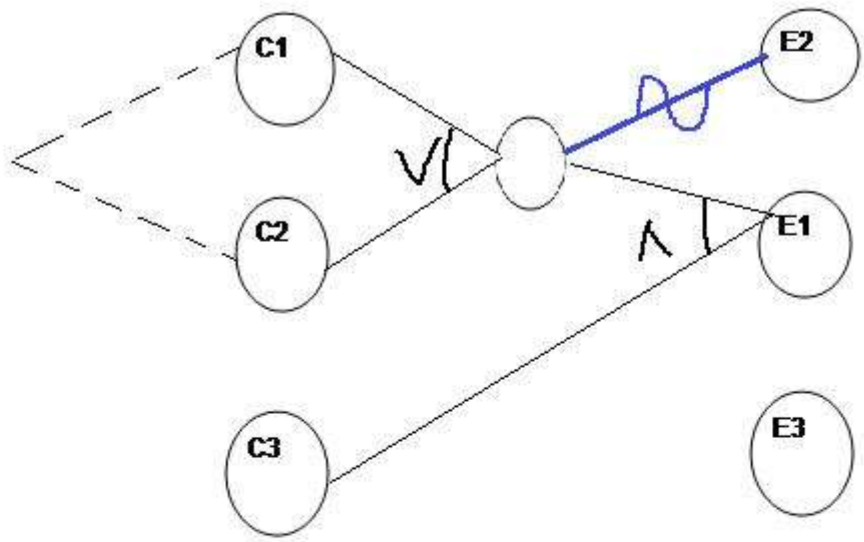
So as per the above diagram, for E1 to be true the condition is

$$(C1 \vee C2) \wedge C3$$

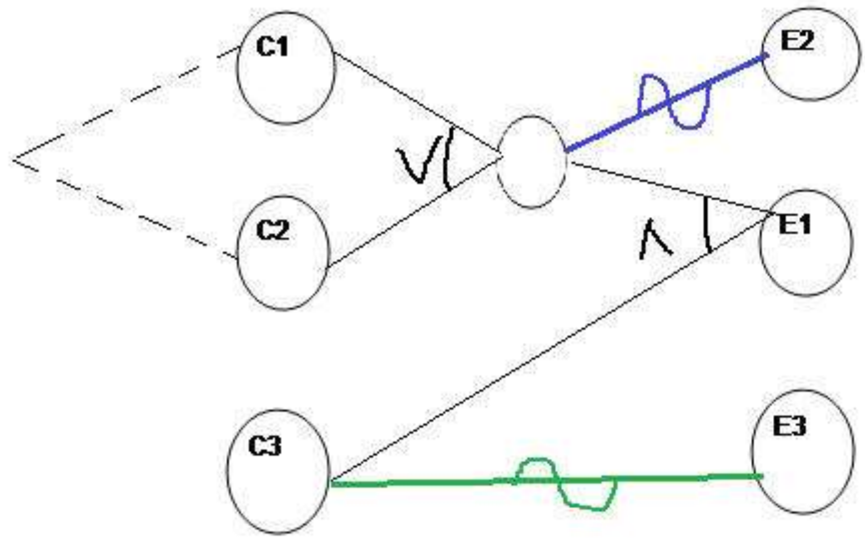
The circle in the middle is just an interpretation of the middle point to make the graph less messy.
There is a third condition where C1 and C2 are mutually exclusive. So the final graph for effect E1 to be true is shown below:



Lets move to Effect E2:
E2 states to print message "X". Message X will be printed when First character is neither A nor B.
Which means Effect E2 will hold true when either C1 OR C2 is invalid. So the graph for Effect E2 is shown as (In blue line)



For Effect E3.
E3 states to print message “Y”. Message Y will be printed when Second character is incorrect.
Which means Effect E3 will hold true when C3 is invalid. So the graph for Effect E3 is shown as (In Green line)



This completes the Cause and Effect graph for the above situation.

Now let’s move to draw the **Decision table based on the above graph**.

Writing Decision table based on Cause and Effect graph

First write down the Causes and Effects in a single column shown below

Actions
C1
C2
C3
E1
E2
E3

Key is the same. Go from bottom to top which means traverse from effect to cause.

Start with Effect E1. For E1 to be true, the condition is: $(C1 \vee C2) \wedge C3$.
Here we are representing True as **1** and False as **0**

First put Effect E1 as True in the next column as

Actions	
C1	
C2	
C3	
E1	1
E2	
E3	

Now for E1 to be “1” (true), we have the below two conditions –
C1 AND C3 will be true
C2 AND C3 will be true

Actions		
C1	1	
C2		1
C3	1	1
E1	1	1
E2		
E3		

For E2 to be True, either C1 or C2 has to be false shown as

Actions				
C1	1		0	
C2		1		0
C3	1	1	0	1
E1	1	1		
E2			1	1
E3				

For E3 to be true, C3 should be false.

Actions						
C1	1		0		1	
C2		1		0		1
C3	1	1	0	1		
E1	1	1				
E2			1	1		
E3					1	1

So it's done. Let's complete the graph by adding 0 in the blank column and including the test case identifier.

Actions	TC1	TC2	TC3	TC4	TC5	TC6
C1	1	0	0	0	1	0
C2	0	1	0	0	0	1
C3	1	1	0	1	0	0
E1	1	1	0	0	0	0
E2	0	0	1	1	0	0
E3	0	0	0	0	1	1

Writing Test cases from the decision table

I am writing a sample test case for test case 1 (TC1) and Test Case 2 (TC2).

TC ID	TC Name	Description	Steps	Expected result
TC1	TC1_FileUpdate Scenario1	Validate that system updates the file when first character is A and second character is a digit.	1. Open the application. 2. Enter first character as "A" 3. Enter second character as a digit	File is updated.
TC2	TC2_FileUpdate Scenario2	Validate that system updates the file when first character is B and second character is a digit.	1. Open the application. 2. Enter first character as "B" 3. Enter second character as a digit	File is updated.

In a similar fashion, you can create other test cases.

and the test case for TC3 to TC6:

TC3
TC3_Print_Msg_X

validate that system print msg X when first character is neither A nor B

1. Open the application
2. Enter 1st character as J
3. Enter 2nd character as h

print msg X

-=-=-=-=-=-=
TC4
TC4_Print_Msg_X

validate that system print msg X when first character is neither A nor B

1. Open the application
2. Enter 1st character as J
3. Enter 2nd character as 8

print msg X

-=-=-=-=-=-=
TC5
TC5_Print_Msg_Y
validate that system print msg Y when 2nd character is a not digit

1. Open the application
2. Enter 1st character as A
3. Enter 2nd character as w

print msg Y

-=-=-=-=-=-=
TC6
TC6_Print_Msg_Y
validate that system print msg Y when 2nd character is a not digit

1. Open the application
2. Enter 1st character as B
3. Enter 2nd character as w

print msg Y